

PLANET: A Multi-objective Graph Neural Network Model for Protein–Ligand Binding Affinity Prediction

Xiangying Zhang,[#] Haotian Gao,[#] Haojie Wang, Zhihang Chen, Zhe Zhang, Xinchong Chen, Yan Li, Yifei Qi,^{*} and Renxiao Wang^{*}



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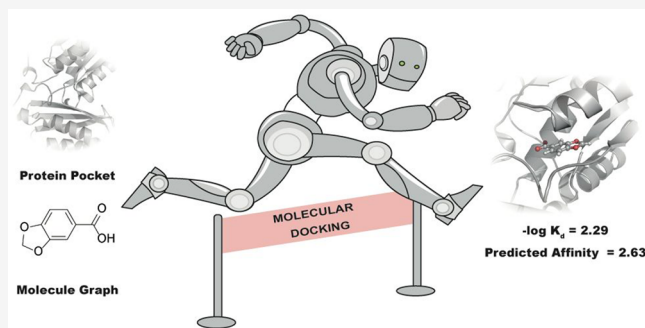


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ABSTRACT: Predicting protein–ligand binding affinity is a central issue in drug design. Various deep learning models have been published in recent years, where many of them rely on 3D protein–ligand complex structures as input and tend to focus on the single task of reproducing binding affinity. In this study, we have developed a graph neural network model called PLANET (Protein–Ligand Affinity prediction NETwork). This model takes the graph-represented 3D structure of the binding pocket on the target protein and the 2D chemical structure of the ligand molecule as input. It was trained through a multi-objective process with three related tasks, including deriving the protein–ligand binding affinity, protein–ligand contact map, and ligand distance matrix. Besides the protein–ligand complexes with known binding affinity data retrieved from the PDBbind database, a large number of non-binder decoys were also added to the training data for deriving the final model of PLANET. When tested on the CASF-2016 benchmark, PLANET exhibited a scoring power comparable to the best result yielded by other deep learning models as well as a reasonable ranking power and docking power. In virtual screening trials conducted on the DUD-E benchmark, PLANET's performance was notably better than several deep learning and machine learning models. As on the LIT-PCBA benchmark, PLANET achieved comparable accuracy as the conventional docking program Glide, but it only spent less than 1% of Glide's computation time to finish the same job because PLANET did not need exhaustive conformational sampling. Considering the decent accuracy and efficiency of PLANET in binding affinity prediction, it may become a useful tool for conducting large-scale virtual screening.



1. INTRODUCTION

Binding affinity to the chosen target protein is a key feature for evaluating designed ligand molecules in drug discovery. Although protein–ligand binding affinity can be measured by various experimental assays, such assays are relatively expensive and often troublesome to set up. Besides, availability of the relevant protein and ligand samples is also an important issue. In contrast, computational methods are efficient, cost-effective, and do not need samples to work. Notably, the accuracy of free energy perturbation methods in binding affinity prediction gets close to the level of experimental errors in certain scenarios.^{1–4} However, application of such methods to high-throughput jobs, such as virtual screening, remains impractical. Combination of molecular docking and scoring function can achieve a balance between accuracy and efficiency, which is the common choice for jobs in reality.⁵ The accuracy of traditional scoring functions, such as X-Score⁶ and AutoDock Vina,⁷ is largely limited by the simple mathematical forms adopted by them.^{8,9} Machine learning methods without predefined formulas can be more flexible to fit the training data and obtain higher accuracy than traditional scoring functions.¹⁰ Δ_{Vina} RF_{20v}¹¹ a random-forest

(RF)-based model, achieved the best metrics in almost all tests in the CASF-2016 benchmark.¹² Machine learning models require feature engineering to transform the raw input to feature descriptors according to predefined rules, which need domain knowledge and can directly affect model performance, preventing end-to-end learning.

In contrast, deep learning models can make predictions based on the most related features automatically extracted from raw input and deep learning models have thus become a research hotspot in binding affinity prediction.^{13,14} Some of those deep learning models do not rely on three-dimensional (3D) structures, while others do rely on 3D structures as inputs. Models in the first category^{15–21} extract features from one-dimensional (1D) or two-dimensional (2D) structural inputs,

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such as protein sequences, SMILES strings, or 2D structural graphs. However, protein–ligand interactions occur in the 3D space, and it is uncertain whether this level of information can be sufficiently degraded into lower dimensions and is thereby learned by this category of models. Models based on 3D structures directly take complex structures as inputs, and one of the most common methods uses a 3D-convolutional neural network (CNN)^{22–31} to extract features from binding sites that have been partitioned into small cubic grids, in which different channels encode structural information such as the positions of different types of atoms. Graph-neural-network (GNN)-based models^{32–35} have also been proposed for predicting binding affinity. Tested on the CASF-2016 benchmark,¹² most of these models can achieve higher scoring power than conventional scoring functions. In addition to scoring power, screening power is also essential for binding affinity prediction models; that is, they should accurately rank binders and non-binders. The screening power determines whether these models can be used in virtual screening, which is one of the most important application scenarios for binding affinity prediction models. Unfortunately, the screening power of many models is less desirable or even being ignored,^{36–38} limiting their value in practice.

To improve the screening capacity of binding affinity prediction models, we have developed PLANET (i.e., Protein–Ligand Affinity prediction NETWORK). PLANET is essentially a GNN model that takes the 3D structure of the binding pocket on the target protein and the 2D structural graph of the ligand molecule as inputs. PLANET was trained through a multi-objective process as multi-objective training has been proven useful for improving the performance and generalization of binding affinity prediction models.³⁹ The three main objectives in our model training included deriving a protein–ligand contact map, ligand distance matrix, and protein–ligand binding affinity. Inspired by the “pairwise interaction” prediction enabled in the sequence-based MONN model,³⁹ the task of deriving the protein–ligand contact map helps PLANET capture protein–ligand interactions from the input structures, and the task of deriving the ligand distance matrix helps PLANET capture 3D features from the 2D structural graph of the ligand molecule. To establish these objectives, besides the commonly used PDBbind data set, artificially generated non-binder decoys⁴⁰ were also added to the model training process. Our PLANET model was evaluated on the popular CASF-2016 benchmark in terms of scoring power, ranking power, and docking power. Moreover, its performance in virtual screening was tested on the DUD-E⁴¹ and LIT-PCBA⁴² data sets, where it exhibited a level of accuracy comparable to that of Glide in only a tiny fraction of the computational time consumed by Glide.

2. METHODS

2.1. Model Architecture. Our PLANET model has three major objectives:

- (1) Prediction of protein–ligand binding affinity. PLANET relies on the 3D structure of the binding pocket on a given target protein and the 2D molecular graph, i.e., chemical structure, of a given ligand molecule to predict the binding affinity between them. It makes a prediction by integrating the contributions from pocket residues and ligand atoms that are involved in interactions. It is of course more straightforward to retrieve protein–ligand interactions from a 3D protein–ligand complex structure, just like the

method employed by Zhu et al. in their work.⁴³ However, the input used by PLANET does not provide such information directly. This difficulty requires PLANET to predict protein–ligand interactions and results in the second objective of PLANET.

- (2) Derivation of the protein–ligand contact map. Here, the protein–ligand contact map is basically a matrix indicating the non-covalent interactions between each pocket residue and each ligand atom. During the supervised training process, PLANET was trained using real protein–ligand interactions observed in crystal structures as labels and learned to predict the protein–ligand contact map based on its input. The predicted protein–ligand contact map is helpful for rescaling the estimated partial energy contribution in the task of binding affinity prediction.
- (3) Derivation of the ligand distance matrix. The ligand distance matrix includes the distance of all atom pairs within the ligand structure. Derivation of this matrix is an auxiliary task to help PLANET extract 3D features from the input 2D molecular graph.

In order to fulfill these three training objectives, PLANET comprises several modules, including protein feature extraction, ligand feature extraction, protein–ligand communication, ligand conformation prediction, pairwise protein–ligand interaction prediction, and affinity prediction (Figure 1).

Protein Feature Encoding and Extraction Module. The structure of a protein pocket is represented as an implicit graph in 3D space, in which each node denotes one pocket residue. The protein feature extraction module mainly consists of the Equivariant Graph Convolutional Layer (EGCL). Our implementation of EGCL is modified from the original Equivariant Graph Neural Network (EGNN) described by Satorras et al.⁴⁴ EGNN was chosen by us because it has proven to be translationally, rotationally, and reflectionally equivariant and computationally more efficient than other models that require spherical harmonics. Given a protein–ligand complex, the ligand centroid of mass is calculated to define a sphere with a radius of 12.0 Å, within which all residues, except for non-natural residues, are considered as pocket residues and defined as nodes in the implicit graph. The initial feature for each node $\{\mathbf{r}_i^{\text{ini}} \in \mathbb{R}^{20}\}_{i=1}^{N_r}$ is the corresponding vector in the BLOSUM62 matrix⁴⁵ and the location for each node $\{x_i \in \mathbb{R}^3\}_{i=1}^{N_r}$ is the coordinate of the α -carbon in the corresponding pocket residue. N_r denotes the number of pocket residues in the graph. Then, the initial features are transformed through an embedding layer (eq 1)

$$\mathbf{r}_i^0 = f(\mathbf{W}^{\text{ini}} \mathbf{r}_i^{\text{ini}})$$
$$f(x) = \max(0, x) + 0.1 \min(0, x) \quad (1)$$

where f represents a LeakyReLU activation function and f shares the same definition hereafter. $\mathbf{W}^{\text{ini}}$ represents a single linear layer with learnable parameters that transforms the input features to a latent space of 300 dimensions. Hereafter, $\mathbf{W}$ shares the same definition, if not specified.

All nodes are updated through EGCL for several iterations. To reduce the number of parameters, the corresponding layers in all updating iterations share the same parameters (eq 2)

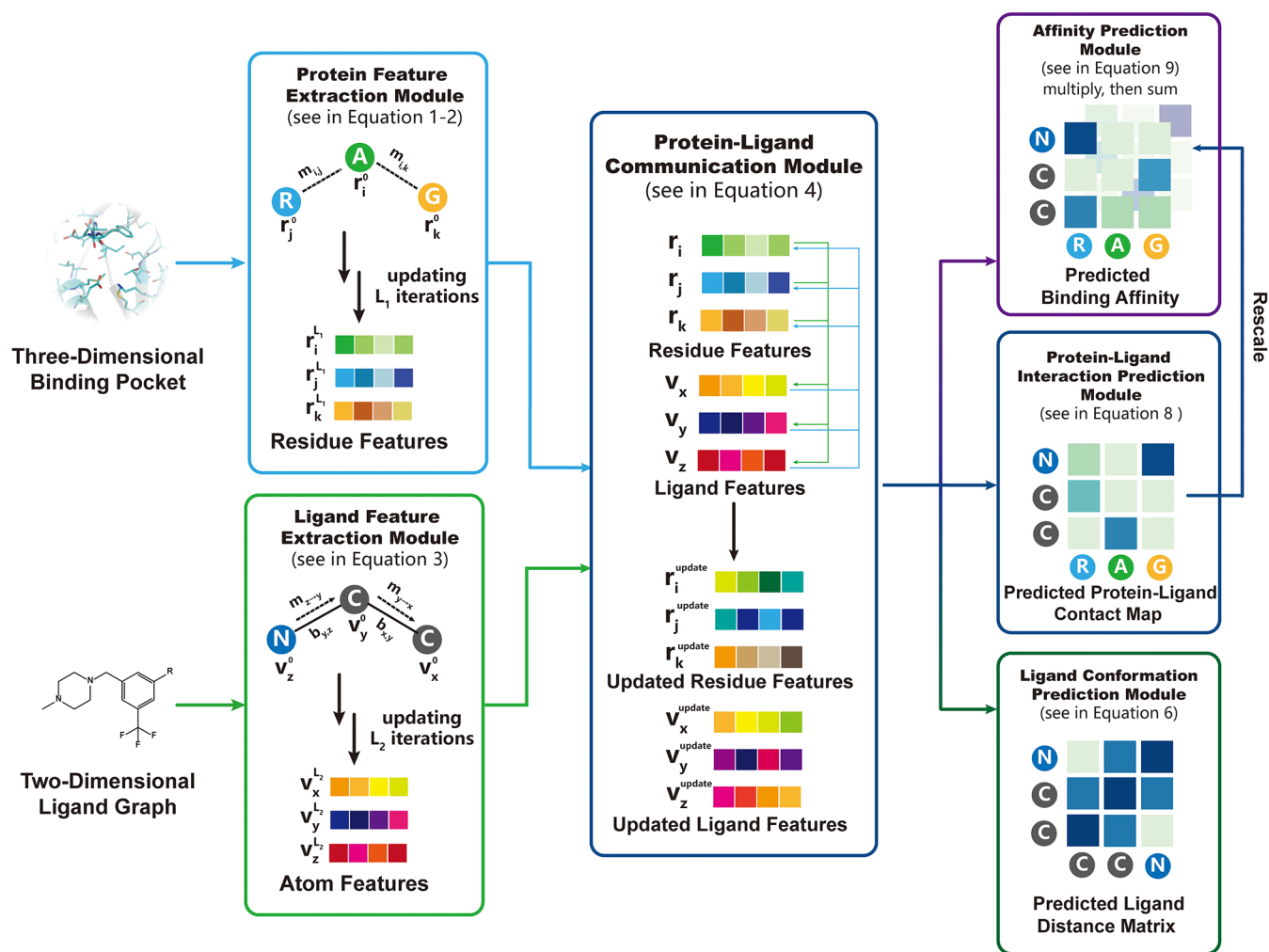


Figure 1. Illustration of the basic framework of PLANET. PLANET implements modules that extract features from the input 3D structure of the binding pocket and the 2D graph of the ligand molecule, respectively. These features are updated by a protein–ligand communication module in a cross-attention manner. The updated features are then utilized to predict protein–ligand binding affinity, protein–ligand contact map, and ligand distance matrix. Note that protein–ligand binding affinity is predicted by summing up energy contributions, which are rescaled by the predicted protein–ligand contact map.

$$\begin{aligned}
 \mathbf{m}_{i,j} &= \phi_e[r_i^l, r_j^l, \|x_i - x_j\|^2] \\
 e_{i,j} &= \phi_{\text{inf}} \mathbf{m}_{i,j} \\
 \mathbf{m}_i &= \sum_{j \neq i} e_{i,j} \mathbf{m}_{i,j} \\
 \mathbf{r}_i^{l+1} &= \phi_h[\mathbf{r}_i^l, \mathbf{m}_i]
 \end{aligned} \quad (2)$$

where ϕ_e denotes two consecutive linear layers with two LeakyReLU activation functions and ϕ_h denotes two consecutive linear layers with one LeakyReLU activation function. ϕ_{inf} denotes one linear layer with one sigmoid activation function to normalize the output to (0, 1) as the relation between the two nodes. $[\cdot, \cdot]$ represents the concatenation of vectors. After L_1 update iterations (here, $L_1 = 3$), the features of all pocket residues ($\mathbf{r}_i^l$) are obtained.

Ligand Feature Encoding and Extraction Module. The structure of a ligand is represented as a directed graph in 2D space, where each node denotes an atom and each edge denotes a covalent bond between the two corresponding atoms. The initial features are one-hot encoded descriptors of atoms

($\mathbf{v}_x \in \mathbb{R}^{27}_{i=1}^{N_v}$ and bonds $\{b_{x,y} \in \mathbb{R}^6\}$ (see Table S1 in the Supporting Information)). The ligand feature extraction module is modified from the graph encoder module in JT-VAE⁴⁶ by introducing a multi-head attention mechanism to better describe the contributions of different kinds of neighbors. Similar to the protein feature extraction module, layers share the same parameters in all update iterations. After L_2 update iterations (here, $L_2 = 10$), the features for all atoms ($\mathbf{v}_x^l$) are obtained (eq 3)

$$\begin{aligned}
 \mathbf{m}_{x \rightarrow y}^l &= f \left(\mathbf{w}_1 \mathbf{v}_x + \mathbf{w}_2 \mathbf{b}_{x,y} + \mathbf{w}_3 \frac{1}{|N(x) - 1|} \sum_{z \in N(x) \setminus y} \hat{\mathbf{m}}_{z \rightarrow x}^{l-1} \right) \\
 \hat{\mathbf{m}}_{z \rightarrow x}^{l-1} &= \frac{1}{K} \sum_{k=1}^K \alpha_{z \rightarrow x}^{k,l-1} \mathbf{w}_v^k \mathbf{m}_{z \rightarrow x}^{l-1} \\
 \alpha_{z \rightarrow x}^{k,l-1} &= \text{softmax} \left(\frac{\mathbf{W}_Q^k \mathbf{m}_{x \rightarrow y}^{l-1} \cdot \mathbf{W}_K^k \mathbf{m}_{z \rightarrow x}^{l-1}}{\sqrt{h_1}} \right)
 \end{aligned}$$

$$\mathbf{v}_y^{L_2} = f\left(\mathbf{w}_1^{\text{out}} \mathbf{v}_y + \sum_{x \in N(y)} \mathbf{w}_2^{\text{out}} \mathbf{m}_{x \rightarrow y}^{L_2}\right) \quad (3)$$

where $|N(x)|$ denotes the number of neighbors of atom x and $N(x) \setminus y$ denotes all neighbors of atom x except for atom y . Dot $(\cdot)$ denotes the inner product of the matrix. K denotes the number of heads in the multi-head attention mechanism. h_1 is the dimension of latent space (here, $h_1 = 300$). $\hat{\mathbf{m}}_{z \rightarrow x}^0$ is zero-initialized.

Protein–Ligand Communication Module. In the aforementioned modules, protein and ligand features are updated in two independent graphs. To facilitate message exchange between protein and ligand features, a protein–ligand message exchange module was designed. The relation between each pair of pocket residue node and ligand atom node is derived in a cross-attention manner and served as weight in the following feature update (eq 4)

$$\begin{aligned} \alpha_{i,j} &= \sigma(\mathbf{W}^{\text{att}}(\mathbf{w}_r^{\text{r}} \mathbf{r}_i^{L_1} \otimes \mathbf{W}^{\text{v}} \mathbf{v}_j^{L_2})) \\ \mathbf{r}_i^{\text{update}} &= \frac{1}{N_v} \sum_{j=1}^{N_v} \alpha_{i,j} \mathbf{w}_u^{\text{v}} \mathbf{v}_j^{L_2} + \mathbf{r}_i^{L_1} \\ \mathbf{v}_j^{\text{update}} &= \frac{1}{N_r} \sum_{i=1}^{N_r} \alpha_{i,j} \mathbf{w}_u^{\text{r}} \mathbf{r}_i^{L_1} + \mathbf{v}_j^{L_2} \end{aligned} \quad (4)$$

where N_v denotes the number of atoms in the molecule, σ denotes a sigmoid activation function, and $\otimes$ denotes element-wise production. For simplicity, the updated residue features ($\mathbf{r}_i^{\text{update}}$) and atom features ($\mathbf{v}_j^{\text{update}}$) are abbreviated as $\mathbf{r}_i$ and $\mathbf{v}_j$, respectively.

Ligand Conformation Prediction Module. This module predicts how atoms interact with each other in 3D space. The obtained atom features ($\mathbf{v}_x$) are transformed to a suitable space to predict the interaction between the two atoms ($\hat{c}_{x,y}^{\text{lig}}$). Distances (in angstroms) between each pair of atoms are calculated to set labels for the ligand distance matrix according to the following criterion (eq 5):

$$c_{x,y}^{\text{lig}} = \begin{cases} 1, & d_{x,y}^{\text{atom}} \leq 4.0 \text{ \AA} \\ 0, & d_{x,y}^{\text{atom}} > 4.0 \text{ \AA} \end{cases} \quad (5)$$

Binary cross entropy loss between the real label ($c_{x,y}^{\text{lig}}$) and the predicted value ($\hat{c}_{x,y}^{\text{lig}}$) is computed as the loss of the ligand conformation prediction module (eq 6)

$$\begin{aligned} \mathbf{v}_x^{\text{contact}} &= \phi_{\text{lig}} \mathbf{v}_x \\ \hat{c}_{x,y}^{\text{lig}} &= \sigma(\mathbf{W}_{\text{lig_contact}}(\mathbf{v}_x^{\text{contact}} \otimes \mathbf{v}_y^{\text{contact}})) \\ l_{x,y}^{\text{lig}} &= -\hat{c}_{x,y}^{\text{lig}} \times \ln \hat{c}_{x,y}^{\text{lig}} - (1 - \hat{c}_{x,y}^{\text{lig}}) \times \ln(1 - \hat{c}_{x,y}^{\text{lig}}) \\ L^{\text{lig}} &= \frac{1}{N_v} \sum_{x,y} l_{x,y}^{\text{lig}} \end{aligned} \quad (6)$$

where $\mathbf{W}_{\text{lig_contact}}$ transforms the hidden features to a one-dimensional vector and ϕ_{lig} is a single linear layer followed by a LeakyReLU activation function.

Protein–Ligand Interaction Prediction Module. The final residue features ($\mathbf{r}_i$) and atom features ($\mathbf{v}_j$) are transformed to predict the interaction ($\hat{c}_{i,j}$) between each residue–atom pair. Labels for the protein–ligand contact map can be calculated

from the crystal structure according to the following rule: if the distance of atom i to any atom in residue j is less than 4.0 Å, the label ($c_{i,j}$) for atom i and residue j is set to 1; else, the label is set to 0 (eq 7):

$$c_{i,j} = \begin{cases} 1, & d_{i,j} \leq 4.0 \text{ \AA} \\ 0, & d_{i,j} > 4.0 \text{ \AA} \end{cases} \quad (7)$$

For a protein–decoy pair, all interaction labels are set to 0, indicating that such a protein and decoy molecule cannot form a stable complex. Binary cross entropy loss between the real label ($c_{i,j}$) and predicted value ($\hat{c}_{i,j}$) is computed as the loss (L^{pl}) of the protein–ligand interaction prediction module (eq 8)

$$\begin{aligned} \hat{c}_{i,j} &= \sigma(\mathbf{W}_{\text{pl}}(\phi_p^1 \mathbf{r}_i \otimes \phi_l^1 \mathbf{v}_j)) \\ l_{i,j}^{\text{pl}} &= -c_{i,j} \times \ln(\hat{c}_{i,j}) - (1 - c_{i,j}) \times \ln(1 - \hat{c}_{i,j}) \\ L^{\text{pl}} &= \frac{1}{N_v} \sum_{i,j} l_{i,j}^{\text{pl}} \end{aligned} \quad (8)$$

where $\mathbf{W}_{\text{pl}}$ transforms the hidden features to a one-dimensional vector. ϕ_p^1 and ϕ_l^1 are two consecutive linear layers with two LeakyReLU activation functions.

Binding Affinity Prediction Module. According to the principles of physical chemistry, there is a linear relationship between the protein–ligand binding free energy and the negative logarithm of the protein–ligand dissociation constant, i.e., $-\log K_d$ or $\text{p}K_d$. We assume that the final residue features $\mathbf{r}_i$ and atom features $\mathbf{v}_j$ can be transformed to predict the partial energy contribution. It is reasonable to assume that, in the whole protein–ligand complex, only the pocket residues and ligand atoms in direct interaction contribute to binding affinity. However, the interacting pairs of pocket residues and ligand atoms cannot be directly inferred from the input binding pocket and two-dimensional ligand structure. To tackle this problem, the predicted protein–ligand interaction $\hat{c}_{i,j}$ given by the protein–ligand interaction prediction module can be viewed as an interacting probability of each residue–atom pair, which is used to reweight the corresponding predicted partial energy contribution. All rescaled partial contributions are summed, and the optimization objective is to minimize the L_2 loss between the predicted affinity and experimental value (eq 9)

$$\begin{aligned} \hat{E}_{i,j}^{\text{pre}} &= f(\mathbf{W}_e(\phi_p^2 \mathbf{r}_i \otimes \phi_l^2 \mathbf{v}_j)) \\ \hat{E}_{i,j} &= \hat{E}_{i,j}^{\text{pre}} \times \hat{c}_{i,j} \\ \widehat{\text{p}K_d} &= \sum_{i,j=1}^{i=N_r, j=N_v} \hat{E}_{i,j} \\ L^{\text{affinity}} &= (\widehat{\text{p}K_d} - \text{p}K_d)^2 \end{aligned} \quad (9)$$

where $\mathbf{W}_e$ transforms the hidden features to a one-dimensional vector. ϕ_p^2 and ϕ_l^2 are two consecutive linear layers with two LeakyReLU activation functions.

Loss Function. The overall loss comes from the three modules described above (eq 10):

$$L^{\text{total}} = \begin{cases} L^{\text{lig}} + L^{\text{pl}} + L^{\text{affinity}} & \text{protein–ligand pair} \\ L^{\text{pl}} & \text{protein–decoy pair} \end{cases} \quad (10)$$

Table 1. Basic Information of Some ML/DL Models and Their Scoring Power Obtained on the CASF-2016 Benchmark

Publish Year	Model	Training Set	Model Input	MAE	RMSE	R_p	Ref.
2023	PLANET	PDBbind general set v.2020 ($N = 15,616$)	graph-represented 3D structure of binding pocket and 2D molecular structure	0.965	1.247	0.824	
2018	TopBP	PDBbind refined set v.2016 ($N = 3767$)	image-like complex structure features	N.A.	1.19 ^a	0.861 ^a	57
2022	DCML	PDBbind refined set v.2016 ($N = 3772$)	Dowker-complex-based representation	N.A.	1.255	0.843	58
2021	PerSpect-GBT	PDBbind refined set v.2016 ($N = 3772$)	graph-represented 3D structure of the protein–ligand complex in various different scales	N.A.	1.244	0.84	59
2021	OPRC-GBT	PDBbind refined set v.2016 ($N = 3772$)	Ollivier persistent Ricci-curvature-based complex representation	N.A.	1.252	0.838	60
2021	IGN	PDBbind general set v.2016 ($N = 8298$)	graph-represented 3D structure of the protein–ligand complex	0.940 ^b	1.22 ^b	0.837 ^b	33
2019	AGL-Score	PDBbind refined set v.2016 ($N = 3772$)	element-specific groups encoded complex structure	N.A.	1.25	0.833	61
2018	K_{DEEP}	PDBbind refined set v.2016 ($N = 3767$)	voxelized represented 3D structure of the protein–ligand complex	N.A.	1.27 ^c	0.82 ^c	27
2019	OnionNet	PDBbind general set v.2016 ($N = 11,906$)	3D structure of the protein–ligand complex described in the number of element-pair-specific contacts	0.984 ^a	1.278 ^a	0.816 ^a	62
2022	MPNN	PDBbind general set v.2016 ($N = 9662$)	graph-represented 3D structure of the protein–ligand complex	N.A.	1.511 ^a	0.813 ^a	34
2021	“Late fusion model”	PDBbind general set v.2016 ^c	graph and voxelized represented 3D structure of the protein–ligand complex	1.044 ^a	1.326 ^a	0.808 ^a	26
2018	RF-Score	PDBbind refined set v.2016 ($N = 3767$)	3D structure of the protein–ligand complex described by the number of element-pair-specific contacts	N.A.	1.39 ^a	0.80 ^a	27
2022	Sfcnn	PDBbind refined set v.2019 ($N = 4100$)	voxelized represented 3D structure of the protein–ligand complex	1.028	1.326	0.793	30
2020	“Dense model”	PDBbind general set v.2016 ^c	voxelized represented 3D structure of the protein–ligand complex	N.A.	1.35 ^a	0.79 ^a	31
2018	Pafnucy	PDBbind general set v.2016 ($N = 11,906$)	voxelized represented 3D structure of the protein–ligand complex	1.13 ^a	1.42 ^a	0.78 ^a	24
2020	“Pair-wise model”	PDBbind refined set v.2018 ($N = 2675$)	3D structure of the protein–ligand complex described in pairwise interactions	N.A.	1.45	0.75	43

^aIt was reported that 290 complexes, including 284 in the standard CASF-2016 test set, were considered in this evaluation. ^bIt was reported that 262 complexes in the CASF-2016 test set were considered in this evaluation and thus the results of this model need to be interpreted with care. ^cThe exact number of samples included in the training set was not mentioned in the reference.

Obviously, a protein–decoy pair does not have the corresponding ligand binding pose and binding affinity, and thus, L_{lig} and $L_{affinity}$ were overlooked in such a case during the training process. It is not necessary to assign the binding affinity of a protein–decoy pair to a small value, since we have set the protein–ligand contact map labels to zero for a protein–decoy pair and therefore the partial energy contribution from each pair of pocket residue and ligand atom will be automatically rescaled to a small value, denoting a low binding affinity. The scales of those three objectives were close to 10.0 at the beginning of training. We had noticed that substantial adjustments to the weights of those objectives only had minor impacts on training outcomes. Thus, all three objectives were assigned equal weights in the final version for the sake of simplicity.

Model Implementation. PLANET was implemented using the PyTorch deep learning framework⁴⁷ along with the RDKit package⁴⁸ for chemical information processing.

2.2. Model Training. Structural preparation. The PDBbind “general set” is essentially a large collection of protein–ligand complexes with available 3D structures from the Protein Data Bank (PDB) and corresponding experimental binding affinity data (i.e., K_d , K_i , and IC_{50}) curated from the literature.⁴⁹ This data set has been a popular choice for deriving deep/machine learning models aiming at protein–ligand binding affinity prediction (Table 1). Our data set for training PLANET was also compiled by taking the PDBbind general set (v.2020), a total of 19,443 protein–ligand complexes, as the starting pool. For each protein–ligand complex in this data set,

structure preparation was performed using the *protein preparation* module in the Schrödinger software (v.2020), including fixing all incomplete residues, setting protonated states, and adding hydrogen atoms. The protonation state at pH 6.5–7.5 was set for the ligand molecule by using the *Epik* module.⁵⁰ The resulting protein structure was saved in the PDB format, and the ligand structure was saved in the SDF format. A number of ligand structures could not pass the sanitize operation in RDKit and thus were removed. As result, a total of 19,386 protein–ligand complexes in the PDBbind general set (v.2020) were successfully processed through the above procedure.

Compilation of the training set. Since our model was later evaluated on the CASF-2016 benchmark, the 285 protein–ligand complexes included in the CASF-2016 test set need to be removed from our data set to avoid overlap. Besides, we have demonstrated in a previous study that the existence of structurally similar samples between the training set and the test set, i.e., “soft overlap”, can lead to an overoptimistic performance of the resulting machine learning model.⁵¹ In order to remove the soft overlap from our training set, we employed the CD-hit software^{52,53} to cluster the entire PDBbind general set (v.2020) by protein sequence with a similarity cutoff of 90%. A protein–ligand complex was regarded as a redundant sample if (i) its protein sequence fell in the same cluster as any complex in the CASF-2016 test set AND (ii) ECFP4 fingerprints of the ligand in this complex and any ligand in the CASF-2016 test set had a Tanimoto coefficient above 0.90. Consequently, a total of 369 complexes were identified as redundant and thus removed

from our data set. After this step, the remaining complexes in each cluster were randomly split into two groups in a ratio of 80:20, where the first group was assigned to the training set and the second group to the validation set. In the end, our training set consisted of 15,616 protein–ligand complexes and the validation set consisted of 3401 protein–ligand complexes.

Augment of non-binder decoys into the training set. The ligand molecule in a protein–ligand complex is of course regarded as a “binder”. In order to derive a model capable of differentiating binders from non-binders, it is a common practice to add some non-binders to the training set. In our work, a certain number of molecules with similar physicochemical properties as the ligand molecule in each protein–ligand complex in the training set were selected from ChEMBL release 29 (<https://www.ebi.ac.uk/chembl/>)⁵⁴ and then added to the training set as non-binder decoys. The workflow of our decoy selection is illustrated in Figure S1 in the Supporting Information. Briefly, molecular properties such as molecule weight, logP, number of hydrogen bond donors, number of hydrogen acceptors, and so on were calculated using the RDKit software for all 2.1 million molecules in ChEMBL 29. For each protein–ligand complex in our training set, up to 100 candidate decoys were randomly selected from ChEMBL 29, and those that met at least six of the following eight criteria were selected out: (1) The difference in molecule weight was less than 10%. (2) The difference in logP is less than 1 unit. (3) The difference in the number of hydrogen bond donors was not larger than 1. (4) The difference in the number of hydrogen bond acceptors was not larger than 1. (5) The difference in the number of chiral centers was not larger than 1. (6) The difference in the number of rings was not larger than 1. (7) The difference in the number of heavy atoms was not larger than 1. (8) The difference in the number of total atoms was less than 10%. Note that the qualified candidates were further examined by docking them into the same binding site as the ligand molecule in the given protein–ligand complex. Here, molecular docking was conducted by using the Glide SP module in the Schrödinger software (v.2020) with default settings, except for “INNERBOX” set to 5 Å and “GLIDE_CONFIG_EFCUT” set to 4.0 kcal/mol. The candidates for which Glide SP failed to output any binding pose were taken as the non-binder decoys for the given protein–ligand complex. This additional molecular docking examination was conducted to reduce the chance that the selected non-binder decoys were actually binders. As result, a total of 161,547 non-binder decoys were added to our training set to augment the 15,616 binders. Each binder was thus accompanied by roughly 10 non-binder decoys.

The training process. A batch size of 16 samples was used during training, where the chance of retrieving a true protein–ligand complex or a protein–decoy pair as one sample was equal. The training process was performed with an Adam optimizer with an initial learning rate of 0.0001 for updating parameters. Model parameters were saved every 5000 batches. The whole training process was completed in 1.5 days on a single NVIDIA P100 GPU card with 16 GB memory. The model with the best performance on the validation set was selected for further evaluation.

2.3. Model Evaluation. Evaluation of scoring power and ranking power. According to the terminology in this field, “scoring power” refers to the ability of a scoring model to predict a binding score, based on valid inputs, that is in linear correlation to experimental protein–ligand binding data. In our work, the scoring power of PLANET was tested on the CASF-2016

benchmark.¹² CASF-2016 has been a popular benchmark for testing scoring functions, including various machine learning models (see Table 1 for some examples). The test set in CASF-2016 consists of 285 diverse protein–ligand complexes with high-quality crystal structures and experimental binding data. Following the settings in CASF-2016, evaluation metrics computed in our scoring power test included the Pearson correlation coefficient (R_p), root-mean-square error (RMSE), and mean absolute error (MAE) between experimental binding data and predicted values. Another related test is the ranking power test. “Ranking power” refers to the ability of a scoring model to correctly rank the known ligands of a common target protein. Besides, ranking power measures how well the scoring model ranks those ligands correctly by their binding data, rather than expecting a linear correlation between the predicted binding data and true values as in the scoring power test. In CASF-2016, the 285 protein–ligand complexes are in fact grouped into 57 clusters, where each cluster contains five complexes formed by the same target protein. For each target protein, the Spearman correlation coefficient (R_s) between experimental binding data and predicted values was computed. The ranking power was the average R_s value across all 57 target proteins. The same bias-corrected and accelerated bootstrapping method used in CASF-2016¹² was employed to obtain the 90% confidence interval of the R_s value. For this task, an ensemble of 10,000 bootstrap samples was considered.

Evaluation of ligand distance matrix and protein–ligand contact map prediction. PLANET was also evaluated regarding its performance in predicting the ligand distance matrix and the protein–ligand contact map. For each task, its performance was measured by the cosine similarity between the true values and predicted ones on the entire CASF-2016 test set. In addition, the area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPRC) were calculated at the atom level, residue level, and pairwise level in the task of protein–ligand contact map prediction.³⁹ To calculate AUROC and AUPRC values at the atom level, ligand atoms with at least one contact with any pocket residue were labeled as positive samples. The predicted value for a ligand atom was the sum of all contacts formed between it and the protein. Cases in which all the ligand atoms were positive samples were excluded from AUROC and AUPRC calculations. The same method was also applied to the calculation of AUROC and AUPRC values at the residue level. In order to testify the predictive power of PLANET further, a random baseline was added to the comparison. This model was designed to generate the ligand distance matrix and protein–ligand contact map randomly by referring to the true ligand structure and protein–ligand complex structure, respectively. For example, assuming that a given protein–ligand complex structure is projected into a protein–ligand contact map consisting of 95% false values (“0”) and 5% true values (“1”), this baseline model would fill up a predicted protein–ligand contact map with 5% values in the range of (0.5, 1.0) and 95% values in the range of (0, 0.5) randomly. This baseline random model was also applied to the entire CASF-2016 test set. Its performance was measured by the same set of evaluation metrics applied to PLANET. The CASF-2106 benchmark also enables a docking power test.¹² Here, “docking power” refers to the ability of a scoring model to identify the true ligand binding pose among decoys. For this purpose, each protein–ligand complex in the CASF-2016 test set is associated with up to 100 decoy ligand binding poses generated by molecular docking in

addition to the native ligand binding pose observed in the crystal complex structure. Our PLANET will produce the same predicted binding score for all ligand binding poses because it takes the 2D ligand structure as input. Therefore, it cannot be subjected to the docking power test directly as other scoring models. However, a protein–ligand contact map can be derived for each ligand binding pose included in CASF-2016, and thus, one can determine which contact map resembles the one predicted by PLANET most closely (also measured by the cosine similarity). In this way, the docking power of PLANET can still be evaluated directly, although PLANET is not trained to predict any explicit ligand binding pose. Following the convention in the docking power test, a complex was recorded as a successful case if the binding pose preferred by the scoring model was close enough ($\text{RMSD} < 2.0 \text{ \AA}$) to the native ligand binding pose. Three sets of evaluations were performed by considering the top one, two, or three binding poses selected by the scoring model.

Evaluation of screening power. In this article, the “screening power” refers to the ability of a scoring model to differentiate binders from non-binders. Two public virtual screening benchmarks, DUD-E⁴¹ and LIT-PCBA,⁴² were employed in our work to test the screening power of PLANET. DUD-E contains 102 target proteins, where each target protein is associated with (i) a protein structure along with a known ligand to define the binding pocket and (ii) relevant binders as well as non-binders in a ratio of roughly 1:50. LIT-PCBA contains 15 target proteins, where each target protein is associated with (i) a certain number of protein–ligand complex structures and (ii) relevant binders as well as non-binders in a ratio of roughly 1:1000. All protein structures and binder and non-binder molecules in these two benchmarks were prepared following the same procedure for processing the PDBbind general set. Note that, because LIT-PCBA provides multiple protein–ligand complexes for each target protein, we only considered the one with the highest resolution in order to save computational cost (see Table S2 in the Supporting Information).

In this test, the AUROC score and enrichment factor (EF) at different thresholds (eq 11) were computed as the measurements of screening power. The enrichment factor was computed with the following formula

$$\text{EF}^\alpha = \frac{\text{NT}^\alpha}{\text{NT}^{\text{total}} \cdot \alpha} \quad (11)$$

where NT^α is the number of binders found among the top α -ranked molecules (e.g., $\alpha = 1\%$) based on the predicted binding affinity. NT^{total} is the total number of binders for the given target in the data set.

To make a comparison to the conventional docking method, the Glide SP module in the Schrödinger software was also tested on the LIT-PCBA benchmark. Glide SP was applied all with default settings. Note that Glide SP failed to generate any docking pose for some molecules on certain target proteins. The docking scores for the molecules in such cases were assigned to zero for the sake of convenience.

2.4. Projection and Visualization of Data Sets. The uniform manifold approximation and projection (UMAP) analysis⁵⁵ was used to illustrate the chemical space covered by the several data sets employed in our work. For this analysis, (i) all true binders and non-binder decoys in our data set and (ii) all binders and 20% of the non-binders in the DUD-E and LIT-PCBA data sets were considered. The ECFP4 fingerprints of all molecules were calculated and hashed to 512 bits using the

RDKit software, resulting in binary vectors with 512 dimensions. Those vectors were then subjected to the incremental principal component analysis reduction⁵⁶ to reduce their dimension to 100. Finally, UMAP analysis was conducted using the *umap-learn* python package with default settings to project the reduced vectors to two dimensions for visualization.

3. RESULTS AND DISCUSSION

3.1. Prediction of Protein–Ligand Binding Affinity.

Among the three trained objectives of PLANET, i.e., protein–ligand contact map, ligand distance matrix, and protein–ligand binding affinity, the primary one is to predict the protein–ligand binding affinity. Following the convention in this field, we first tested the “scoring power” of PLANET, i.e., computing the binding affinity of a protein–ligand complex based on its structure, on the CASF-2016 test set. The correlation between the experimental binding affinity data and the values predicted by PLANET is shown in Figure 2, where the Pearson correlation

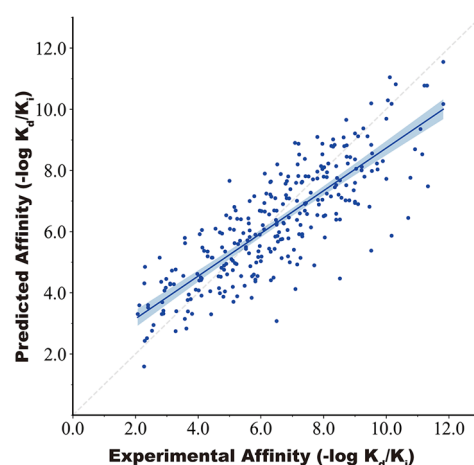


Figure 2. Correlation between the experimental binding affinity data and the predicted values by PLANET on the CASF-2016 test set ($N = 285$; $R_p = 0.824$; $\text{MAE} = 0.965 \text{ log units}$; $\text{RMSE} = 1.247 \text{ log units}$). The solid line is the regression line, where the 95% confidence interval is indicated by the shaded region.

coefficient is 0.824. The basic information on a number of deep/machine learning models published in recent years, including the scoring power obtained on the same test set, is summarized in Table 1. Most of those models produce a Pearson correlation coefficient around 0.80. Thus, PLANET is at least comparable to the best among them in this test.

Here, the readers should be reminded that the real “scoring power” of PLANET may be underrated by these statistical metrics for two reasons. First, all other models included in Table 1 directly take 3D protein–ligand complex structures as input, while PLANET only uses 3D binding pocket structures and 2D ligand structures. Obviously, it is more challenging to predict the protein–ligand binding affinity without the exact information of their binding mode. In fact, the scoring power of some deep learning models that rely on crystal protein–ligand complex structures would decrease considerably if complex structures derived from molecular docking are used instead. In the case of the “fusion model” reported by Jones et al.,²⁶ for example, when complex structures derived from molecular docking were used instead of crystal structures, the R_p and RMSE values of their model dropped from 0.808 and 1.326 log units to 0.712 and

1.871 log units, respectively, on the CASF-2016 test set. We suspect that other deep/machine learning models included in Table 1 would exhibit a lower scoring power if they were subjected to the same treatment. The scoring power test enabled by the CASF-2016 benchmark is certainly a basic test of the essential quality of a protein–ligand interaction scoring function. However, in reality, it is not very meaningful to “predict” the binding affinity of a protein–ligand pair based on their crystal complex structure because it is far more difficult to obtain crystal complex structures than measuring the binding affinity experimentally. From this perspective, our PLANET model is more robust and more efficient for the purpose of protein–ligand binding affinity prediction without the need of obtaining or generating the corresponding protein–ligand complex structures in prior.

Second, during model training, we removed the protein–ligand complexes that are structurally too similar to those in the CASF-2016 test set with a similarity cutoff of 90% applied to the protein and ligand sides. We call such samples “soft overlap” between the training set and the test set. We have demonstrated in our previous study⁵¹ that the existence of such samples in the training set leads to overoptimistic statistical metrics in the scoring power test for machine learning scoring functions. Therefore, we suggested that removing the soft overlap between the training and test sets should be a standard practice for testing a deep/machine learning model. However, none of the models listed in Table 1 actually executed this standard practice in their training process. It is unclear to what extent their statistical metrics (R_p and RMSE) in this scoring power test would decrease if they did so.

Another test enabled by CASF-2016 is the ranking power test. As explained in section 2.3, the so-called ranking power emphasizes correct ranking of the ligand molecules binding to the same target protein by their binding affinity. In our opinion, this test may resemble more closely how a scoring model is applied in reality. In this ranking power test, PLANET yielded an average Spearman correlation coefficient of 0.682 on the 57 target proteins in the CASF-2016 test set. As one can see in Figure 3, the performance of PLANET is better than all other conventional scoring functions included in CASF-2016, except for a random-forest machine learning model $\Delta_{\text{vina}}\text{RF}_{20}$. Note that, for unknown reasons, all other deep learning models listed in Table 1 have not been tested on CASF-2016 in terms of ranking power. Thus, a head-to-head comparison to them is not possible at this time.

3.2. Depiction of Protein–Ligand Interactions. As a special feature in its working mechanism, PLANET predicts binding affinity by integrating only the contributions from the interacting pairs of pocket residues and ligand atoms. This mechanism requires PLANET to derive the protein–ligand contact map at the first place. This, however, is not a straightforward task because the inputs to PLANET from the ligand side are merely 2D molecular graphs. Thus, prediction of the protein–ligand contact map is included as a major objective in our model training. Besides, an auxiliary objective to derive the ligand distance matrix is added to help PLANET extract 3D features from 2D molecular graphs. Such a working mechanism is quite different from many deep learning models listed in Table 1, which typically rely on 3D protein–ligand complex structures for prediction. Importantly, inclusion of protein–ligand contact map and ligand distance matrix predictions also makes our model interpretable, at least to some extents, in terms of

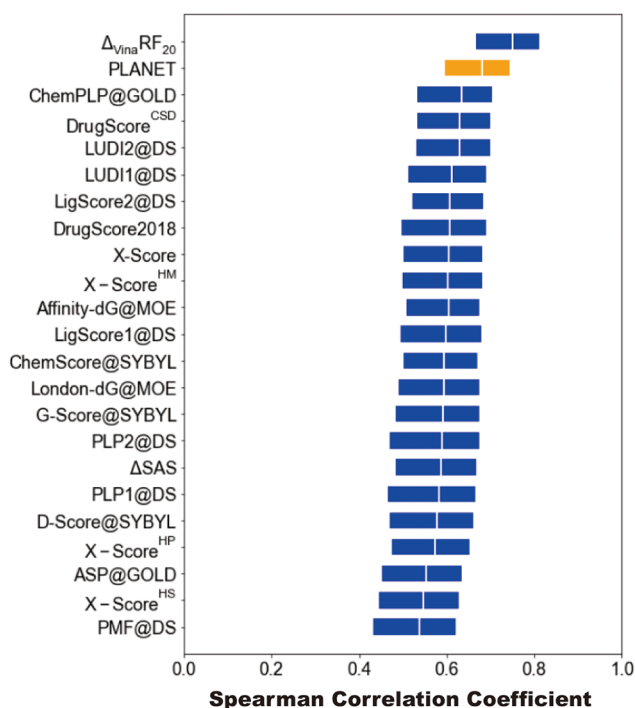


Figure 3. Ranking power of PLANET as obtained on the CASF-2016 test set. The average Spearman correlation coefficient obtained on 57 target proteins is indicated by a short white line, where the bar indicates the 90% confidence interval derived from bootstrapping sampling. Data for other models included in this figure are cited from ref 12.

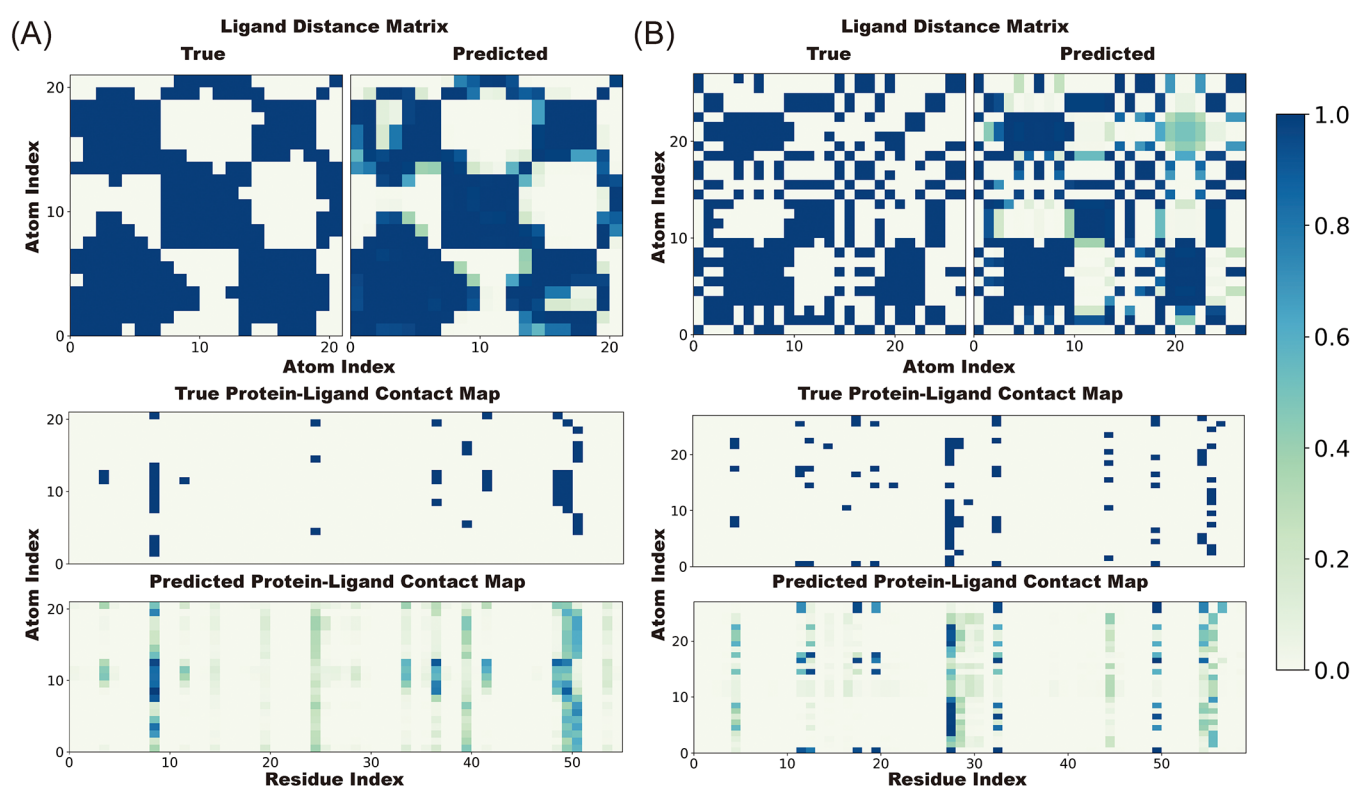
capturing the essential 3D features in protein–ligand interaction.

The quality of the protein–ligand contact map and ligand distance matrix predicted by PLANET is measured by the cosine similarity between the matrix derived from the real structure and the predicted one. The statistical results obtained on the entire CASF-2016 test set are summarized in Table 2. In addition, as prediction of the protein–ligand contact map and ligand distance matrix is essentially a binary classification task, AUROC and AUPRC values at different levels were also calculated and summarized in Table 2. Generally, PLANET is able to predict the ligand distance matrix well with an overall cosine similarity of 0.925 on the CASF-2016 test set. In comparison, the overall cosine similarity given by the baseline model is 0.519 in the same case. Regarding protein–ligand contact map prediction, the overall cosine similarity given by PLANET is 0.583. This result seems to be somewhat less promising. However, considering that the overall cosine similarity given by the baseline model is as low as 0.201 in this case, one can conclude that the accuracy of PLANET in this aspect is still encouraging. Besides, as measured by the AUROC and AUPRC values calculated at different levels, the accuracy of our PLANET is superior to that of the baseline model by a large margin in both ligand distance matrix and protein–ligand contact map prediction (Table 2).

As described in the Methods section, PLANET predicts binding affinity by integrating the contributions from the pocket residues and ligand atoms in direct interaction. However, such information is not indicated explicitly in its input. This requires PLANET to predict the protein–ligand contact map at the first place. The predicted protein–ligand contact map can be considered as the interacting probability of each residue–atom

Table 2. Performance of PLANET and a Baseline Model in Ligand Distance Matrix and Protein–Ligand Contact Map Prediction on the CASF-2016 Test Set

Tasks		AUROC	AUPRC	Cosine Similarity
Performance of the PLANET Model				
Ligand distance matrix		0.975 $\pm$ 0.033	0.961 $\pm$ 0.061	0.925 $\pm$ 0.049
Protein–ligand contact map	Atom-level	0.756 $\pm$ 0.200	0.968 $\pm$ 0.043	0.583 $\pm$ 0.114
	Residue-level	0.964 $\pm$ 0.039	0.944 $\pm$ 0.056	
	Pairwise-level	0.934 $\pm$ 0.040	0.509 $\pm$ 0.170	
Performance of a Baseline Model				
Ligand distance matrix		0.500 $\pm$ 0.017	0.385 $\pm$ 0.131	0.519 $\pm$ 0.105
Protein–ligand contact map	Atom-level	0.517 $\pm$ 0.201	0.911 $\pm$ 0.092	0.201 $\pm$ 0.021
	Residue-level	0.492 $\pm$ 0.083	0.349 $\pm$ 0.112	
	Pairwise-level	0.501 $\pm$ 0.027	0.060 $\pm$ 0.013	

**Figure 4.** Two examples of the ligand distance matrix and protein–ligand contact map correctly predicted by PLANET. (A) PDB entry 3G2Z: experimental $-\log K_i = 2.36$, predicted value = 2.52; (B) PDB entry 3RYJ: experimental $-\log K_i = 7.80$, predicted value = 7.97.

pair, which is then used to rescale its contribution. Therefore, the predictive power of PLANET is rooted in its capability of capturing the essential 3D features in protein–ligand interactions. To provide some explicit examples, the predicted ligand distance matrix and protein–ligand contact map for two selected protein–ligand complexes in the CASF-2016 test set are illustrated in Figure 4. One can see that the predicted matrices largely resemble the ones derived from the real structures, where the predicted binding affinity data are also close to the true values. There are of course some cases where the ligand distance matrix and protein–ligand contact map predicted by PLANET are not correct and the binding affinity data are poorly predicted. Two such examples are provided in Figure S2 in the Supporting Information. All of these examples demonstrate that, unlike some other deep learning models that are essentially black boxes, the outputs produced by PLANET are more interpretable.

As explained in section 2.3, the protein–ligand contact map derived by PLANET does provide an indirect way for evaluating this model in the docking power test with the CASF-2016 benchmark. Here, PLANET's success rates of including the native ligand binding pose among the top one to three candidates are 71.8%, 81.6%, and 87.9%, respectively, which already surpasses a number of conventional scoring functions (Figure 5). This level of performance is in fact encouraging because PLANET is not particularly trained for predicting ligand binding pose. Note that all other deep learning models listed in Table 1 have not been subjected to this docking power test, and thus, a direct comparison with them is again not possible.

This docking power test demonstrates the achievement of our multi-objective training strategy for deriving the PLANET model. Nevertheless, this test also indicates that more accurate prediction of the protein–ligand contact map is desired in order to improve PLANET in this aspect. This is a challenging task for PLANET under its current framework. Protein–ligand contact

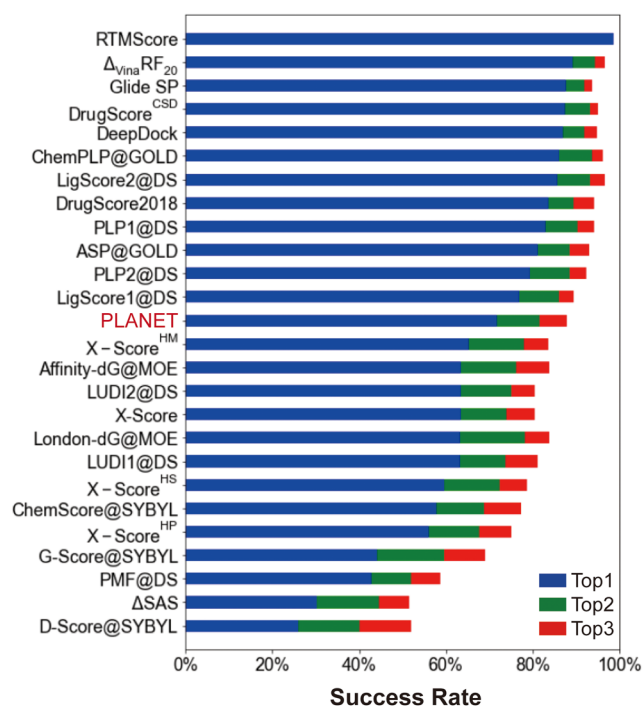


Figure 5. Docking power of PLANET as obtained on the CASF-2016 benchmark, where the success rates of including the native ligand binding pose among the top one to three candidates are illustrated. Here, the data for DeepDock and RTMScore are cited from ref 63 and ref 64, respectively, while the data for all other models are cited from ref 12.

map prediction is assigned as a binary classification task, but the labels in our training set are extremely unbalanced. In fact, the labels for more than 95% of the possible residue–atom pairs are blank. Such unbalanced training data of course presented a troublesome challenge in our model development. Note that the AUROC and AUPRC values computed at the residue level (0.964 and 0.944, respectively) indicate that PLANET captures most of the important residues interacting with the ligand molecule, albeit some false positives do exist among the predictions.

It should be mentioned that accurate prediction of the ligand binding pose by the deep learning model in another framework is possible. For example, DeepDock⁶³ and RTMScore⁶⁴ are also included in the comparison illustrated in Figure 5. One can see that their performance is excellent in this docking power test. However, unfortunately those deep learning models have not

been evaluated in binding affinity prediction. Currently, it still remains as a high-hanging fruit in this field to obtain a true multipurpose deep learning model.

3.3. Performance in Virtual Screening. It should be mentioned again that, although the “scoring power” is an essential aspect of a successful scoring model, in reality scoring models are applied more often to virtual screening tasks, i.e., differentiating true binders from non-binders. In other words, the “screening power” of a scoring model should receive more attention. Several CNN models described in the literature, such as Pafnucy and OnionNet, were shown to achieve higher scoring power on the CASF-2016 test set than traditional scoring functions, but they could hardly distinguish between binders and non-binders in virtual screening trials.³⁶ A common strategy for achieving a good screening power through deep learning is to derive a classification model by using a special training set consisting of binders and non-binders.^{33,65–68} This strategy is of course sound in concept. However, if the training set and the chosen test set are congeneric in nature, e.g., splitting a common mother data set,^{51,69} deep learning models tend to exhibit overoptimistic performance.

In order to avoid this potential caveat in model evaluation, we tested the screening power of PLANET on the DUD-E benchmark directly after its training on the PDBbind data set was accomplished; i.e., no further fine-tuning was attempted to extend its generalizability. The results produced by PLANET, together with those produced by OnionNet, Pafnucy, and several other machine learning models reported by Shen et al.,³⁶ are summarized in Table 3. One can see that the performance of all deep/machine learning models originally trained on the PDBbind data set was much less promising in this test, because (i) the composition of the DUD-E data set is distinctively different from the PDBbind data set and (ii) the goal switches from regression to classification in this scenario. In contrast, our PLANET exhibited a reasonable level of performance here. For example, PLANET (AUROC = 0.746) is superior to Pafnucy (AUROC = 0.631) and OnionNet (AUROC = 0.597), given that all three of these deep learning models were trained on the PDBbind data set and achieved a comparable level of scoring power on the CASF-2016 benchmark.

Very interestingly, the conventional scoring function Glide SP achieved the best performance among all models in terms of AUROC score and EFs. This reflects the fact that, although conventional scoring functions are limited by their simple mathematical architecture in encoding a large, complex body of data, they are able to capture some basic physics in protein–ligand interaction and thus may have a generalized performance

Table 3. Performance of Six Deep Learning/Machine Learning Models as Well as Two Conventional Scoring Functions on the DUD-E Benchmark

Model	AUROC		EF ^{0.5%}	EF ^{1%}	EF ^{5%}
	mean ± std	Median	Mean	Mean	Mean
PLANET	0.746 ± 0.124	0.747	10.581	8.850	5.378
Δ _{vina} RF ₂₀ ^a	0.697 ± 0.102	0.695	9.516	7.995	4.382
RFscore-V4 ^a	0.652 ± 0.108	0.655	4.902	4.521	2.981
NNscore2.0 ^a	0.683 ± 0.098	0.683	4.162	4.022	3.123
Pafnucy ^a	0.631 ± 0.124	0.639	4.241	3.861	2.767
OnionNet ^a	0.597 ± 0.102	0.606	2.840	2.840	2.205
Glide SP ^a	0.767 ± 0.116	0.783	19.389	16.182	7.231
Vina ^b	0.716	N.A.	9.139	7.321	4.444

^aPerformance metrics are cited from ref 36. ^bPerformance metrics are cited from ref 66.

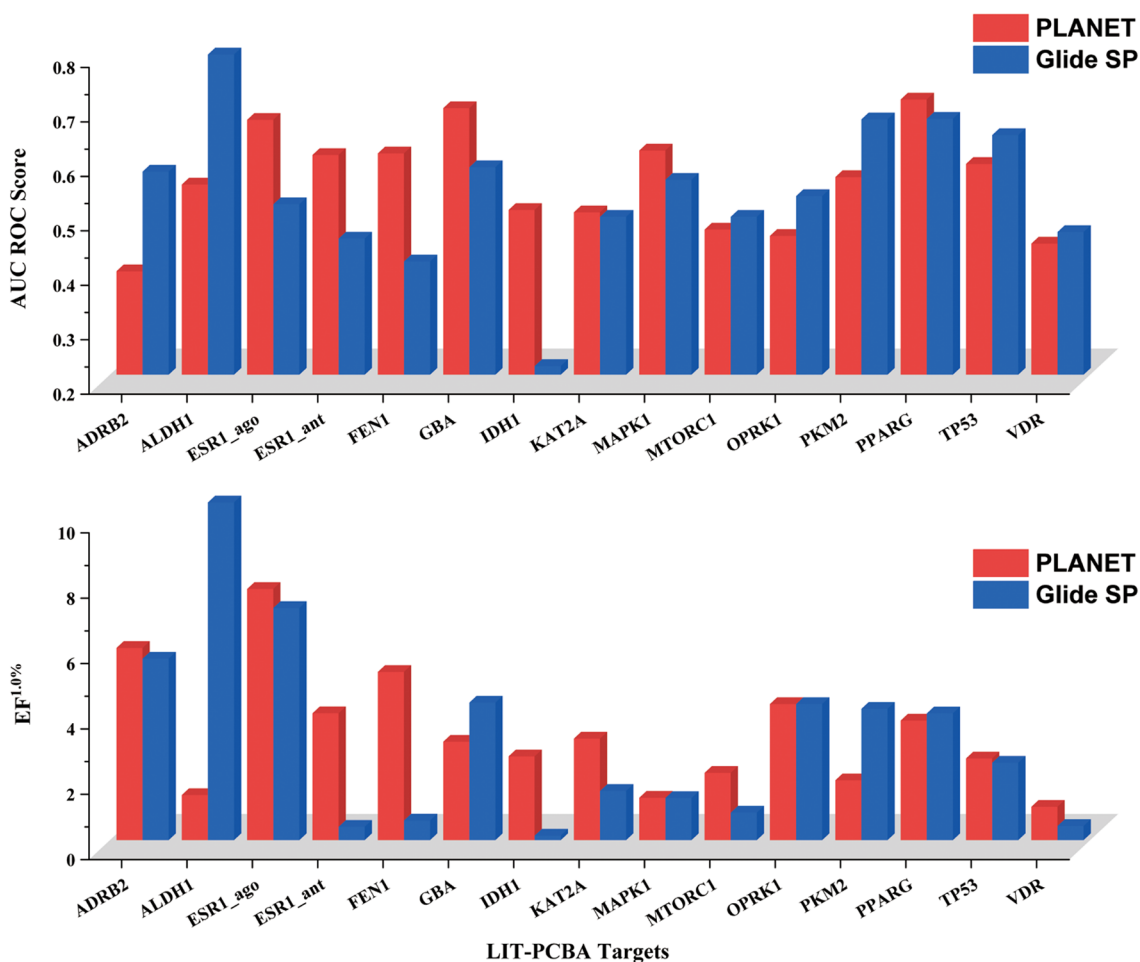


Figure 6. Evaluation results of PLANET (red) and Glide SP (blue) on the LIT-PCBA data set. In terms of AUROC or EF^{1.0%}, no significant difference was observed between the results given by PLANET and Glide SP by paired *t*-test.

in virtual screening tasks. Another piece of evidence supporting this observation comes from the results of $\Delta_{\text{Vina}}\text{RF}_{20}$, which also outperformed other deep/machine learning models in this test. Note that $\Delta_{\text{Vina}}\text{RF}_{20}$ is in fact a combination of the Vina scoring function and a correction term derived through a random forest model.¹¹ That correction term has improved the scoring power of $\Delta_{\text{Vina}}\text{RF}_{20}$. However, as one can see in Table 3, it brings no advantage to the screening power as compared to Vina. In other words, the screening power of $\Delta_{\text{Vina}}\text{RF}_{20}$ is actually established by the Vina scoring function. Although our PLANET model is second to Glide SP on the DUD-E benchmark, it outperformed all other deep/machine learning models. PLANET demonstrated a good level of generalizability even though the goal here has shifted from the original regression task.

Although DUD-E is perhaps the most popular benchmark for evaluating virtual screening models, a few studies have pointed out that the artificially generated decoys included in the DUD-E data set may lead to biased results.^{42,69–72} In order to conduct a more objective evaluation of screening power, we further tested PLANET head-to-head with Glide SP on the recently published LIT-PCBA benchmark.⁴² In this new benchmark, all binders and decoys are experimentally proven according to the information from PubChem BioAssay.⁷³ All 2.7 million molecules are carefully selected through an asymmetric validation embedding procedure. Besides, for some targets included in LIT-PCBA, the number of decoys is roughly 1000 times larger than that of

binders, which better mimics the real situation in virtual screening.

Similar to the case of DUD-E, our PLANET model was directly evaluated on LIT-PCBA without fine-tuning. Glide SP was applied in its standard workflow and parameter setting. The evaluation results are illustrated in Figure 6. More details can be found in Table S3 in the Supporting Information. Here, one can see that LIT-PCBA is indeed more challenging than DUD-E, as the mean AUROC scores of PLANET and Glide SP are 0.556 and 0.536, respectively, both of which are lower than the counterparts obtained on DUD-E. Compared to Glide SP, PLANET produced higher AUROC scores on eight targets, comparable scores on three targets, and lower scores on four targets in LIT-PCBA. Nevertheless, paired *t*-test revealed no significant difference between the results produced by PLANET and Glide SP, i.e., $p = 0.497$ for AUROC scores and $p = 0.845$ for EF^{1.0%}. Thus, the screening power demonstrated by these two models on the LIT-PCBA benchmark is basically on par. Nevertheless, PLANET had a huge technical advantage in terms of efficiency: It took PLANET about 3 h to finish all jobs needed by this test on a single NVIDIA P100 GPU card, while it took Glide SP about 25 days to finish the same test on an Intel XEON W-2245 CPU (16 cores, 3.90 GHz). Perhaps a more appropriate comparison should be made by employing a GPU-implementation of Glide SP, but such an implementation is not available at present. It is obvious that PLANET achieves its computational

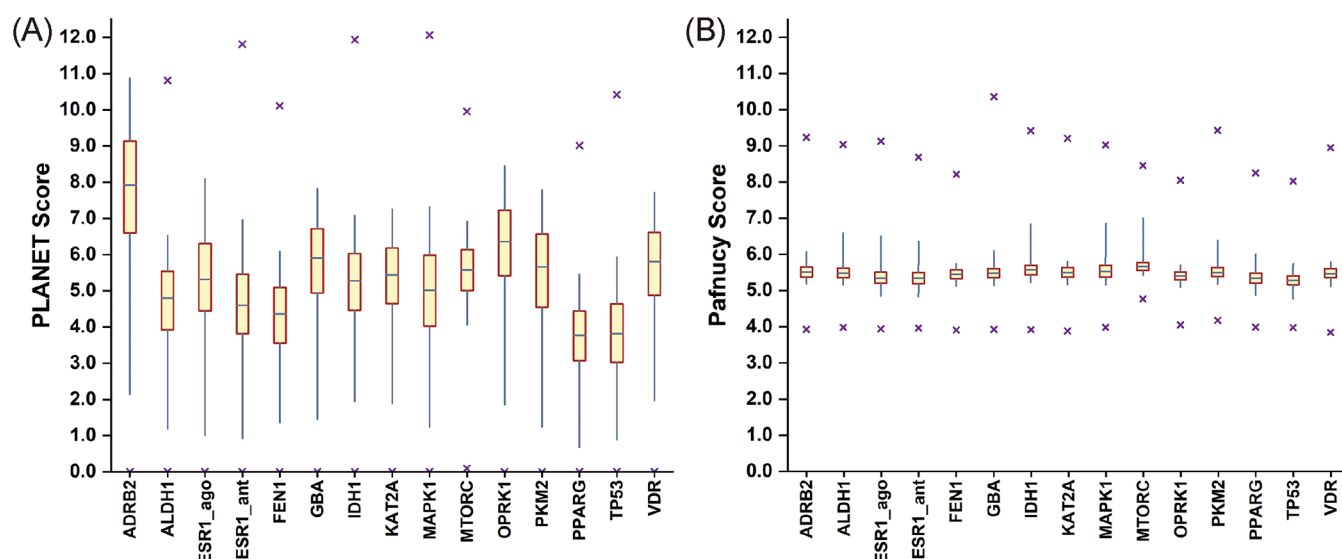


Figure 7. (A) Distribution of the protein–ligand binding affinity data predicted by PLANET on the LIT-PCBA data set. (B) Distribution of the protein–ligand binding affinity data predicted by Pafnucy on the same data set, where the data are cited from ref 78. In both figures, labels on the *x* axis are the name of each target protein. The boxes delimit the first and third quartiles of the predicted values, while the whiskers delimit the 5th and 95th percentiles. The mean value is indicated by the dash in the middle of each box. The minimum and maximum values in each case are indicated by crosses.

Table 4. Scoring Power and Screening Power of PLANET in Several Ablation Studies

Model	Scoring Power as on CASF-2016			Screening Power as on DUD-E			
	MAE	RMSE	R_p	AUROC	EF ^{0.5%}	EF ^{1%}	EF ^{5%}
PLANET	0.965	1.247	0.824	0.746	10.581	8.850	5.378
PLANET _{crystal}	0.924	1.226	0.830	0.661	5.032	4.645	3.132
PLANET _{2D}	1.062	1.412	0.774	0.684	6.482	5.530	3.464
PLANET _{baseline}	1.007	1.297	0.805	0.656	4.373	3.801	2.744

efficiency by skipping the time-consuming molecular docking process and replaces it by prediction of the protein–ligand contact map. Thus, PLANET can hopefully become a practical tool for virtual screening of extremely large compound databases in addition to the current approaches.^{74–77}

We have noticed that Pafnucy, a CNN-based deep learning model, was also tested on the LIT-PCBA data set.⁷⁸ According to the reported results, the screening power of Pafnucy was rather poor on this benchmark. In fact, nearly 90% of the binding affinities predicted by Pafnucy are distributed in a narrow range between 5.0 and 6.5 (in $-\log K_d/K_i$ units) in a basically target-independent manner (Figure 7B). This failure may result from the voxelized representation of the protein binding site by a CNN model, which produces a sparse tensor with extremely high dimensions, and it is in turn difficult to extract useful features from the sparse tensor.⁶⁹ In contrast, PLANET produced binding affinities in a much wider range in a more realistic, target-dependent manner (Figure 7A). This suggests that PLANET does have some “understanding” of protein–ligand interactions and thus makes meaningful predictions.

3.4. Impact of the Training Set on Model Performance.

Besides predicting protein–ligand binding affinity, our PLANET model has additional training objectives including predicting the ligand distance matrix and protein–ligand contact map. Additional training data are thus needed for those objectives. In order to reveal the improvements achieved by using additional training objectives and training sets, we performed several ablation studies as follows: (1) “PLANET_{crystal}”: This model

shared the same three objectives as the full model, but it was trained only on the crystal protein–ligand complex structures in the PDBbind data set, with no decoy molecules added. (2) “PLANET_{2D}”: This model did not include the objective of predicting the ligand distance matrix, but its training set was the same as the full model. (3) “PLANET_{baseline}”: This model did not include the objective of predicting the ligand distance matrix, and it was trained only on the crystal protein–ligand complex structures in the PDBbind data set. The above three additional models were trained in a similar process as the full model. After training, they were tested on the CASF-2016 benchmark to evaluate their scoring power and on the DUD-E benchmark to evaluate their screening power. The evaluation results are summarized in Table 4.

As shown in Table 4, the scoring power of all three additional models was roughly at the same level, i.e., R_p around 0.80 on the CASF-2016 test set. Interestingly, when the decoys were removed from our training set, the resulting model, i.e., PLANET_{crystal}, even achieved a marginally better scoring power (MAE = 0.924 log units) than PLANET itself (MAE = 0.965 log units). This indicates that the PDBbind data set itself is sufficient for training a model with a promising scoring power on CASF-2016, which is in fact a common practice for most of the deep/machine learning models summarized in Table 1. This is however not a very challenging task because the test set in CASF-2016 is also derived from the PDBbind data set, and thus, the test set and the training set are homogeneous in this scenario.

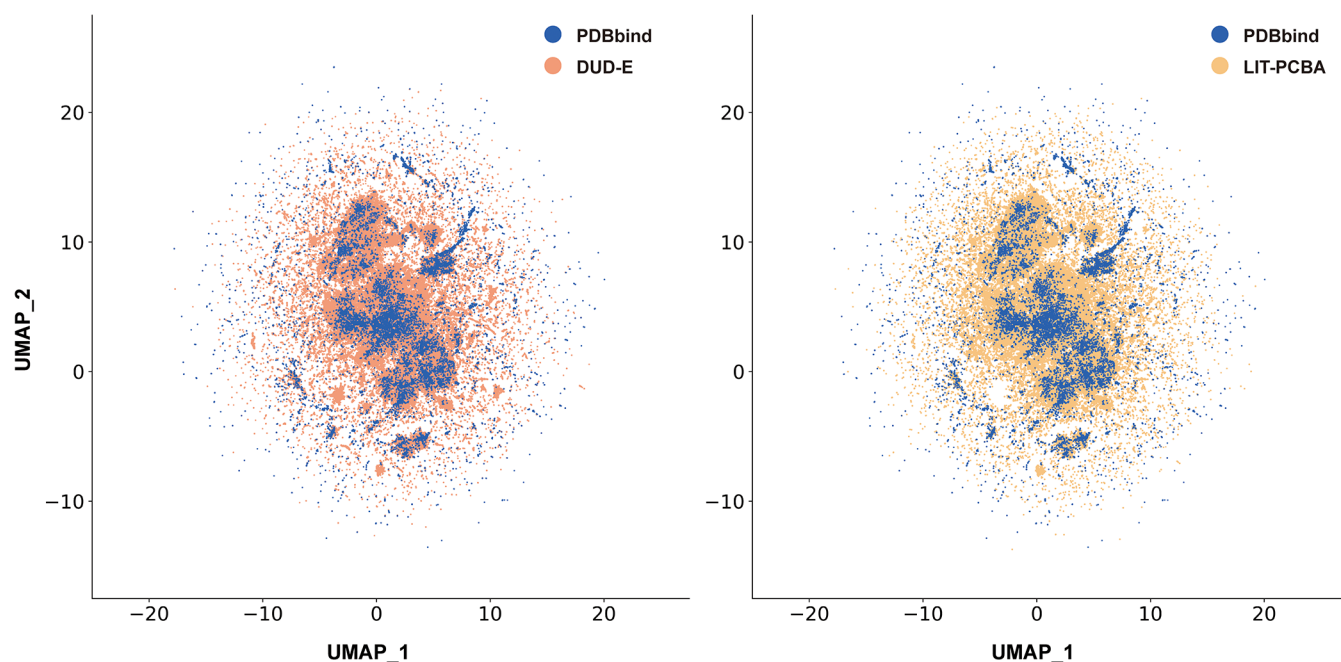


Figure 8. UMAP projections of the molecules in three data sets employed in our work, where blue dots represent the samples in the PDBbind general set (v.2020), salmon dots for those in DUD-E, and orange dots for those in LIT-PCBA.

As it comes to the screening power, a substantial difference is witnessed among the several models. For example, the AUROC score produced by PLANET is 0.746, while the value produced by *PLANET_crystal* is 0.661. This gap between the performance of two models indicates that adding decoy molecules into the training set helps to build a better screening power. Note that the screening power refers to the ability to distinguish between binders and non-binders. The PDBbind data set is formed by known protein–ligand complexes, i.e., all ligand molecules included in this data set are strong or weak binders. Consequently, such a data set is not sufficient for deriving a deep learning model with a good screening power because the model has no chance to learn about any non-binder during the training process. In fact, the strategy of adding decoys into the training set has been applied by other deep learning models published in the literature.⁴⁰ However, our PLANET model has implemented one additional feature to further enhance its screening power, i.e., prediction of the ligand distance matrix. The improved performance of PLANET (AUROC = 0.746) over *PLANET_2D* (AUROC = 0.684) indicates that the task of predicting the ligand distance matrix encourages PLANET to extract features from 2D molecular graphs and in turn helps with its screening power. If disabling the task of ligand distance matrix prediction and also removing decoys from the training set, it is not surprising to observe that the resulting model, i.e., *PLANET_baseline*, exhibited the lowest screening power (AUROC = 0.656).

Apparently, there is still much room for PLANET to improve its screening power. For example, its performance on either DUD-E or LIT-PCBA has not surpassed Glide SP. This does not fulfill the expectation that a deep learning model should be superior to conventional methods. This is probably because the total number of protein–ligand complexes in our training set is still too small to support training a robust deep learning model. The chemical space covered by the ligand molecules from the PDBbind data set is rather limited as compared to that covered

by the molecules included in the DUD-E or LIT-PCBA data set. As proof, the results of our UMAP analysis of those several data sets are given in Figure 8. One can see that, while the binders and non-binders in the DUD-E or LIT-PCBA data set are distributed sparsely over a wide space, the ligand molecules in the PDBbind general set (v.2020) tend to concentrate at certain regions. This observation is consistent with the claim by several other reports that the structural diversity of the ligand molecules in the Protein Data Bank is somewhat limited.^{34,79,80} Adding the decoys selected from ChEMBL29 to our training set greatly expanded its covered chemical space to roughly the same level of DUD-E or LIT-PCBA (see Figure S3 in the Supporting Information). As discussed above, this treatment helps PLANET to improve its screening power (Table 4). However, we have observed that this treatment does not help PLANET much to improve its prediction of the protein–ligand contact map. It is because those added decoys introduce basically all-zero labels in a large number to the training set with respect to the protein–ligand contact map, which makes the training set even more unbalanced for this task. What an “ideal” training set for our multi-objective model should be is of course a matter for debate. Inclusion of significantly more protein–ligand complexes with low binding affinity seems to be necessary for obtaining a more balanced training set. Such samples, however, are relatively insufficient from publicly available resources at present.

We have demonstrated in this work that the multi-objective training strategy and the training set augmented with selected decoys can improve the screening power and generalizability of our PLANET model. Nevertheless, it seems that the insufficient diversity represented by the current training set sets a fundamental limit to its performance. In particular, crystal structures of protein–ligand complexes are not straightforward to obtain, so the number of such structures will not grow dramatically in the near future. From this perspective, more efficient experimental or computational methods for obtaining reliable protein–ligand complex structures are still desired.

Alternatively, more capable machine learning models that can utilize small data sets in training may also provide a solution to the problem of protein–ligand binding affinity prediction.

4. CONCLUSION

In this work, we have developed a GNN-based model, called PLANET (i.e., Protein–Ligand Affinity prediction NETWORK), which takes the 3D structure of the protein binding pocket and the 2D ligand structure as input to predict protein–ligand binding affinity. The essential feature of PLANET is that it is trained as a multi-objective model, where binding affinity prediction is based on a protein–ligand contact map predicted in prior. As mentioned in the Introduction section, the objective of predicting the protein–ligand contact map is inspired by the MONN model.³⁹ However, there are three major differences in our method. First, MONN requires a full-length protein sequence as input, while PLANET requires the 3D structure of the protein binding pocket as input. Second, MONN uses the predicted interactions as attention weight to update ligand and residue features in a subsequent dual-attention network, while PLANET uses the predicted protein–ligand interactions directly to rescale the contribution from each residue–atom pair. This allows PLANET to integrate the non-covalent interactions between protein and ligand. Third, PLANET also includes prediction of ligand distance matrix as another training objective, which helps to extract 3D-related ligand features from the two-dimensional graph input. Unlike some other black-box-style deep learning models, both the protein–ligand contact map and the ligand distance matrix make the outputs of PLANET more interpretable.

Evaluations conducted on the popular CASF-2016 benchmark showed that the scoring power of PLANET was among the top as compared to other deep learning models. Besides, PLANET also exhibited reasonable ranking power and docking power, indicating its potential to become a useful multipurpose model. The value of PLANET is further demonstrated in virtual screening trials conducted on the DUD-E and LIT-PCBA data sets. Here, PLANET outperformed other deep/machine learning models in terms of AUROC scores and EF values. Its accuracy in those tests was generally comparable to that of the established docking method Glide. However, its computational speed is faster by at least 2 orders of magnitude because it does not need the time-consuming molecular docking process. With its decent accuracy and efficiency, we expect PLANET to become a practical tool for conducting large-scale virtual screening tasks in real drug discovery efforts.

■ ASSOCIATED CONTENT

Data Availability Statement

Source codes of PLANET, as well as user manual and demo examples, are available at <https://github.com/ComputArtCMCG/PLANET>.

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.3c00253>.

Descriptions of the nodes and edges in 2D molecule graph construction (Table S1); PDB codes of the protein structures used in the virtual screening trial with the LIT-PCBA data set (Table S2); target-by-target comparison of PLANET and Glide SP on the LIT-PCBA data set (Table S3); workflow of selecting the non-binder decoys in our training set (Figure S1); two examples where the ligand

distance matrix and the protein–ligand contact map are not accurately predicted by PLANET (Figure S2); UMAP projection of the true binders and non-binder decoys included in our training set (Figure S3) (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Yifei Qi – Department of Medicinal Chemistry, School of Pharmacy, Fudan University, Shanghai 201203, People's Republic of China; orcid.org/0000-0003-2853-7910; Email: yfq@fudan.edu.cn

Renxiao Wang – Department of Medicinal Chemistry, School of Pharmacy, Fudan University, Shanghai 201203, People's Republic of China; orcid.org/0000-0003-0485-0259; Email: wangrx@fudan.edu.cn

Authors

Xiangying Zhang – Department of Medicinal Chemistry, School of Pharmacy, Fudan University, Shanghai 201203, People's Republic of China

Haotian Gao – Department of Medicinal Chemistry, School of Pharmacy, Fudan University, Shanghai 201203, People's Republic of China; orcid.org/0000-0002-6153-6058

Haojie Wang – Department of Medicinal Chemistry, School of Pharmacy, Fudan University, Shanghai 201203, People's Republic of China

Zhihang Chen – Department of Medicinal Chemistry, School of Pharmacy, Fudan University, Shanghai 201203, People's Republic of China

Zhe Zhang – Department of Medicinal Chemistry, School of Pharmacy, Fudan University, Shanghai 201203, People's Republic of China

Xinchong Chen – Department of Medicinal Chemistry, School of Pharmacy, Fudan University, Shanghai 201203, People's Republic of China

Yan Li – Department of Medicinal Chemistry, School of Pharmacy, Fudan University, Shanghai 201203, People's Republic of China; orcid.org/0000-0002-8259-2470

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.jcim.3c00253>

Author Contributions

*X. Zhang and H. Gao contributed equally to this work.

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Notes

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